

EXP 5

PROGRAMS BASED ON DECISION MAKING AND KNOWLEDGE REPRESENTATION

1. Family Relationships

Problem: Define family relations and answer grandparent queries.

Program:

```
% Facts
parent(john, mary).
parent(mary, susan).
parent(john, mike).

% Rule
grandparent(X, Y) :-
    parent(X, Z),
    parent(Z, Y).
```

Query:

```
?- grandparent(john, susan).
```

Output:

```
true.
```

2. Animal Classification

Problem: Identify whether an animal is a mammal based on its features.

Program:

```
% Facts
has_hair(tiger).
gives_milk(tiger).

% Rule
mammal(X) :-
    has_hair(X),
    gives_milk(X).
```

Query:

```
?- mammal(tiger).
```

Output:

```
true.
```

3. Simple Arithmetic Knowledge Base

Problem: Implement simple addition using Prolog.

Program:

```
% Rule
add(X, Y, Z) :-
    Z is X + Y.
```

Query:

```
?- add(5, 7, Z).
```

Output:

```
Z = 12.
```
